RMSC4102 Research Project

A Toolkit for Robust Risk Assessment Using F-Divergences

Ma Ting Tin

Department of Statistics, The Chinese University of Hong Kong

April 5, 2022

Agenda

Distance between Distributions

- Recall: For $x, y \in \mathbb{R}$, we can "measure" their distance by $|x - y|$.

Question

- Suppose we model X by f (with CDF v).

Given another distributions g, how can we "measure" distance between f and g, i.e. the discrepancy between distribution?

Natural Guess: Construct a function F to quantify distance between two point and consider its expected value under v.

$$d(f, g) = \mathbb{E}_v \left\{ F\Big(f(X), g(X) \Big) \right\}$$

Naive candidates of F:

1. $F(x, y) = |x - y|$
2. $F(x, y) = (x - y)^2$

Remark: The idea is indeed informal. But for the moment, it serves as a good start to introduce the notion in such way.

Motivation of the Main Problem

After some analysis, one proposed $X \sim f$ for modelling the **loss of this year** of a company. However, it is nearly impossible for us to well-specify the true DGP f_0.

Natural Question

Suppose X is non-negative valued R.V. and

- $f \neq f_0$, i.e. the model does not well specify the true DGP.
- $d(f, f_0) < \epsilon$, i.e. the model is **not perfect but not bad**.

Then how will serious could it affect our analysis on the quantity of interest $\theta := \mathbb{E}(X)$, say the value of

$$\boxed{L := \sup_{g: d(f,g) < \epsilon} \mathbb{E}_g(X),}$$

which is an important quantity in field of **Worst Case Analysis**.

Indeed, there exists quite a number of literature investigating the properties and inference of L.

Main Contribution of the paper

There are several difficulties on handling the mentioned problem.

1. The quantity L is hard to estimate due to its complicated structure.
2. The uncertainty set $\{g : d(f, g) < \epsilon\}$ is hard to interpret.
3. The value of L depends heavily on choice of function F, while there is no absolute answer to the choice of F.

The author proposed a flexible toolkit to solve the above problems at once. In particular, under the toolkit, the following could be done

1. **(Estimation of L):** L could be estimated easily through numerical method.
2. **(Guideline for choice of F):** Provide way to construct F such that the uncertainty set being interpretable.

Agenda

F-divergence

We formally provide F-divergence as one of the possible way for measuring distance between distributions as follow:

Definition (F-divergences)

1. Let $X \in \mathcal{L}^1$ defined on the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ with support $\mathcal{S} := \Omega_X$.
2. Let v and f be the CDF and PDF of X respectively.
3. Let η with density g be another distribution with support $\mathcal{S}$ being absolutely continuous w.r.t. v.

Then for convex F with $F(1) = 0$, F-divergence between v and η is defined as

$$D_F(\eta|v) = \int_{\mathcal{S}} F\left(\frac{g(x)}{f(x)}\right) f(x)dx$$

Remark

F-Divergence may NOT a metric even in L^2 function space as in general,

$$D_F(\eta|v) = \int_{\mathcal{S}} F\left(\frac{g(x)}{f(x)}\right) f(x)dx \neq \int_{\mathcal{S}} F\left(\frac{f(x)}{g(x)}\right) g(x)dx = D_F(v|\eta).$$

Concerns on Choice of F

Example 1 (Popular Choice of F)

1. **Kullback-Leibler divergence:** $F_{KL}(y) = F(y) = y\log(y) - y + 1$.
2. **Polynomial Divergence:** For $p > 1$, $F_p(y) = [y^p - p(y-1) - 1]/[p(p-1)]$.

It is clear to see F_{KL} and F_p grows linearly and exponentially respectively. Hence, we can imagine the uncertainty set induced are different, i.e.

$$\{\eta : D_{F_{KL}}(\eta|v) < \kappa\} \neq \{\eta : D_{F_p}(\eta|v) < \kappa\}$$

Takeaway

- Choice of F reflect attitude of user against loss, which is hard to quantify.
- Improper Choice of F might induces uncertainty set with overly few or many alternative model, which make the worst case analysis being too pessimistic or optimistic.

New Parameterization of F-divergence

Key Idea

Instead of chosing F directly, the author propose to by pass to problem to chosing the transformed derivative of F.

Definition (Alternative parameterization of F-divergence)

Let $H : \mathbb{R} \to \mathbb{R}$ be continuous, strictly increasing with $\lim_{y\to\infty} H(y) = \infty$ and $H(0) = 0$. The associated F is defined by

$$F(y) := \int_1^y H\Big(\log(z)\Big)dz,$$

where $F(0) := \lim_{y\downarrow 0} F(y)$. For simplicity, we denote it as F is induced by H.

Example 2 (Examples of Induced F)

1. KL-divergence: $F_{KL}(y)$ is induced by $H(y) = y$.
2. Polynomial divergence: $F_p(y)$ is induced by $H(y) = [e^{(p-1)y} - 1]/(p-1)$.

Properties of the proposed parameterization

Lemma 1 (Properties of Induced F)

Let F be induced by H. Then F satisifes

1. Convexity
2. $F \in C^1$
3. $\lim_{y \to \infty} F(y)/y = \infty$

Also, $F(1) = F'(1) = 0$ and F attains its minimum 0 at 1. Furthermore if $H \in C^1$ with $H'(0) = 1$, $F \in C^2$ with $F''(1) = 1$.

There are several advantages to define induce F through G.

Remarks

1. **(No loss in generality)** Any sufficiently smooth, convex F be induced by some H.
2. **(Closed-form inverse)** The derivative of induced F have closed-form inverse, which gives theoretical guarantee and makes the simulation possible.

Worst-Case Analysis

Given the Random Variable $X \sim v$. Suppose we are interested in the quantity

$$\boxed{W(v, F, \kappa) := \sup_{\eta : D_F(\eta|v) \leq \kappa} \mathbb{E}_\eta(X) = \sup_{\eta : D_F(\eta|v) \leq \kappa} \int_{\mathcal{S}} x g_\eta(x) dx}$$

- As Supremum is defined as worst case, X should be interpreted as loss.

There is no guarantee that $\boldsymbol{W(v, F, \kappa)} < \infty$ in general. The author provide **sufficient condition** for $W(v, F, \kappa) < \infty$ together with expression for worst case.

To be specific, we give the following definition:

Definition (Worst-Case Density)

g^{wc} is the worst case density with induced probability measure η^{wc} if

1. $D_F(\eta^{wc}|v) \leq \kappa$
2. $\mathbb{E}_{\eta^{wc}}(X) = \int_{\mathcal{S}} x g^{wc}(x) dx = W(v, F, \kappa)$

Sufficient Conditions for Wosrt Case Analysis

Proposition 1

Let $L := \lim_{y\to\infty} H(y) \in [-\infty, 0)$. Assume for all $\alpha \in [0,\infty)$,

$$\int_{\mathcal{S}} |x| \exp\left(H^{-1}(\alpha|x|)\right) f(x)dx < \infty$$

Then we have

1. **(Finiteness of W)** $W(v, F, \kappa) < \infty$ for all $\kappa \in \mathbb{R}^+$.
2. **(Explicit form for g^{wc})** $\exists \kappa_{\max} \in (0,\infty]$ s.t. whenever $\kappa \in (0, \kappa_{\max})$, $\exists \alpha_1^{wc} \in \mathbb{R}, \alpha_2^{wc} \in \mathbb{R}^+$ s.t.
$$g^{wc}(x) = \mathbb{1}_{(\alpha_1^{wc}+\alpha_2^{wc}x>L)} \exp\left(H^{-1}(\alpha_1^{wc} + \alpha_2^{wc}x)\right) f(x).$$

Interpretation and remark for the parameters

1. α_2^{wc} can be interpreted in two ways
   - Generalized exponential tilting parameter that increase the weight on large realizations of X compared with the reference model.
   - Unique parameter satisfying $D_F(\eta^{wc}|v) = \kappa$.
2. α_1^{wc} is chosen such that $\int_{\mathcal{S}} g = 1$

Interpretation with Practical Sufficient Condition

The previous theorem is useful as it implies the following:

Interpretation of the Theorem

1. The worst case analysis is not meaningless, i.e. it does not tell that the worst case is inifinitely bad $(W(v, F, \kappa) = \infty)$.
2. It provides closed form for g^{wc}, which get rid of the complicated convex analyais in most of the related literature.

However, it is still hard to verify the sufficient condition. The author then propose a lemma that provides conditions that are easier to be verified.

Lemma 2 (Sufficient condition for Proposition 1)

Suppose for all $\alpha \in (0, \infty)$, $\exists \epsilon > 0$ s.t.

$$\limsup_{|x| \to \infty} |x|^{2+\epsilon} \exp\left(H^{-1}(\alpha|x|)\right) f(x) < \infty$$

then $\int_{\mathcal{S}} |x| \exp\left(H^{-1}(\alpha|x|)\right) f(x) dx < \infty$ for all α. Also for $\widehat{v}$ with density $\widehat{f}$, if $\limsup_{|x| \to \infty} \widehat{f}(x)/f(x) < \infty$, then $\widehat{v}$ also satisfies the above sufficient condition for boundedness of the integral.

Motivation of the problem

In the last section, we provide conditions to ensure finiteness of $W(v, F, \kappa)$. The question we are going to answer in this question is as follow

Problem to be answered

1. What does the abstract set $\mathcal{D}_F := \{\eta : D_F(\eta|v) < \infty\}$ contains?
2. Standard Routine in existing literature

$$\boxed{\text{Given a choice of } F} \quad \Rightarrow \quad \boxed{\text{Investigate properties of } \mathcal{D}_{\mathcal{F}}}$$

Then is it possible that we can go in the reverse way, i.e.

$$\boxed{\text{Given desired properties } (\star) \text{ of } \mathcal{D}_{\mathcal{F}}} \quad \Rightarrow \quad \boxed{\text{Tailor-made } F \text{ for } (\star)}$$

The answer is partially affirmative by using the proposed toolkit.

Importance of $\mathcal{D}_F$

$\mathcal{D}_F$ is sensitive to the choice of F and the reference model.

Example 3 ($\mathcal{D}_\mathcal{F}$ with General reference model v)

- $\forall \kappa > 0, \exists \eta$ s.t. $\mathbb{E}_\eta(|X|) = \infty$ with $D_{F_{KL}}(\eta|v) \leq \kappa$ (hence imply $\eta \in \mathcal{D}_{F_{KL}}$).
- $\mathcal{D}_{F_p} = \{\eta : \mathbb{E}_\eta(|X|^k) < \infty$ for all k and $\sup_{|x|\to\infty} |g_\eta(x)/f(x)| < \infty\}$

$\boxed{F_{KL}} \Rightarrow \boxed{\text{"Large" } D_{F_{KL}}} \Rightarrow \boxed{\text{Inappropriate } F} \Leftarrow \boxed{\text{"Small" } D_{F_p}} \Leftarrow \boxed{F_p}$

Interpretation of the examples

- F_{KL} induces divergence ball with too extreme model ($W(v, F_{KL}, \kappa) = \infty$).
- F_p induces divergence ball with too restricted model.

Takeaway: Both of the popular choice of F, F_{KL} and F_p are inappropriate for general reference model, which motivate us to consider alternative construction of F.

Matching between F and v

Main idea: Control number of moments that alternative models in divergence ball possess through suitable matching between H and reference model.

Proposition 2

Let η be distribution on $\mathcal{S}$ being absolutely continuous w.r.t v with density g. Suppose that $F(0) < \infty$, $\int_{\mathcal{S}\cap[-k,k]} F(g(x)/f(x))f(x)dx < \infty,\ \forall k \in \mathbb{R}^+$ and $\limsup_{|x|\to\infty} g(x) < 1$. Assume for $\theta > 1$,

1. $$\limsup_{|x|\to\infty} \frac{H\big(-\log f(x)\big)}{|x|^\theta} \mathbb{1}_{\mathcal{S}(x)} < \infty$$
2. $$\mathbb{E}_\eta(X) = \int_{\mathcal{S}} |x|^\theta g(x)dx < \infty$$

Then $D_F(\eta|v) < \infty$.

Remark

Proposition 2 tells the function H should be tailored such that one can intentionally include models with moments up to order θ in the uncertainty sets.

Matching between F and v

By proposition 2, we know that models up to order θ are contained in the uncertainty set, but it does not guarantee that model with order greater than θ are excluded from the uncertainty set. Fortunately, it is true under some regularity conditions

Proposition 3

Let $\theta > 1$ and $t \in (1, \theta)$. Assume that $F(0) < \infty$ and $\exists \widehat{x} \in (0, \infty), \widetilde{y} \in (1, \infty)$ s.t. $(\widehat{x}, \infty) \subset \mathcal{S}$ and H is continuously differentiable on $(\widetilde{y}, \infty)$. Moreover, assume that $\limsup_{y \to \infty} H'(y)/H(y) < \infty$. Also if

1. $\liminf_{x\to\infty} \frac{H\left(-\log f(x) - (t+1)\log x - c\right)}{x^t} > 0$ for all $c \in \mathbb{R}^+$
2. $\liminf_{x\to\infty} x^{t+1} g(x) > 0$
3. $\lim_{x\to\infty} \frac{g(x)}{f(x)} = \infty$

then $D_F(\eta|v) = \infty$.

Proposition 2 and 3 together gives the following implication: Under some regularity conditions, the uncertainty set is given by

$$\boxed{\mathcal{D}_F = \mathcal{D}_F(\theta) := \{\eta : \mathbb{E}_\eta(|X|^\theta) < \infty \text{ and } \mathbb{E}_\eta(|X|^{\theta+\epsilon}) = \infty \text{ for all } \epsilon > 0\}}$$

Summary of Toolkit

To summarize, the Toolkit proposed

1. New Parameterization of F-divergence by H

to derive the class of theory for

2. Characterization of Worst-case distribution

in sense of

1. Providing sets of sufficient conditions to ensure finiteness of W.
2. Providing **Computable Explicit form** of worst-case distribution

$$g^{wc}(x) = \mathbb{1}_{(\alpha_1^{wc}+\alpha_2^{wc}x>L)} \exp\left(H^{-1}(\alpha_1^{wc}+\alpha_2^{wc}x)\right) f(x) := T(\alpha_1^{wc}+\alpha_2^{wc}x) f(x)$$

and finally provide criteria of chosing H such that

3. The induced uncertainty set is interpretable

in sense that we can set the upperbound for the moment of distribution in the uncertainty set.

Point Estimation

However, we only specified the existence of $\alpha_1^{wc}, \alpha_2^{wc}$ in explicit form of g^{wc} without knowing it exact value. We then discuss its estimation and statistical properties of W.

Theorem (Point Estimation)

Suppose we have $x_1, \cdots, x_m$ being independent realizations of X. By Importance Sampling, we have

$$\mathbb{E}_{\eta^{wc}}[X] = \mathbb{E}_v[XT(\alpha_1^{wc} + \alpha_2^{wc} X)] \approx \boxed{\frac{1}{m}\sum_{j=1}^{m} x_j T(\widehat{\alpha}_1^m + \widehat{\alpha}_2^m x_j)}$$

hence reducing the problem into estimation of α_1^{wc} and α_2^{wc}. The author consider the following estimating equations: (implying consistency of $\widehat{\alpha}_1^m, \widehat{\alpha}_2^m$)

- $(D_F(\eta^{wc}|v) = \kappa)$: $\boxed{\frac{1}{m}\sum_{j=1}^{m} F(T(\widehat{\alpha}_1^m + \widehat{\alpha}_2^m x_j)) = \kappa}$
- $(\int_{\mathcal{S}} g^{wc}(x)dx = 1)$: $\boxed{\frac{1}{m}\sum_{j=1}^{m} T(\widehat{\alpha}_1^m + \widehat{\alpha}_2^m x_j) = 1}$

Interval Estimation

We then study how to construct confidence interval for the worst case expectation.

Theorem (Interval Estimation)

Under technical conditions (refer to theorem 2 in chapter 8 of Manski (1988)), we have

$$\sqrt{m}\left(\frac{1}{m}\sum_{j=1}^{m} x_j T(\widehat{\alpha}_1^m + \widehat{\alpha}_2^m x_j) - \mathbb{E}_{\eta^{wc}}[X]\right) \xrightarrow{d} N(0, v^2),$$

where $v^2 = w^T (B^{-1})^T \Sigma B^{-1} w$. Define $\mathcal{U} = \alpha_1^{wc} + \alpha_2^{wc} X$. Then Σ is the covariance matrix of

$$\begin{pmatrix} T(U) \\ F(T(U)) \\ XT(U) \end{pmatrix}$$

under nominal model v, then

$$B = \begin{pmatrix} \mathbb{E}_v[T'(U)] & \mathbb{E}_v[F'(T(U))T'(U)] & 0 \\ \mathbb{E}_v[T'(U)X] & \mathbb{E}_v[F'(T(U))T'(U)X] & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad w = \begin{pmatrix} \mathbb{E}_v[XT'(U)] \\ \mathbb{E}_v[X^2 T'(U)] \\ -1 \end{pmatrix}$$

Implementation

The summary of the procedure is as follow

1. Numerically evaluate $F(y) = \int_1^y H(\log(z))dz$ through spline interpolation.
2. Set initial guess as $\widehat{\alpha}_1^{wc} = a_1$ and $\widehat{\alpha}_2^{wc} = a_2$, where a_1 and a_2 satisfies empirical version of $a_1 = -a_2\mathbb{E}_v(X)$ and $a_2^2\mathbb{V}ar_v(X) = 2\kappa$.
3. Numerically solve the estimating equations by using initial guess a_1, a_2 and
$$\frac{1}{m}\sum_{j=1}^{m} F(T(\widehat{\alpha}_1^m + \widehat{\alpha}_2^m x_j)) = \kappa \quad \text{and} \quad \frac{1}{m}\sum_{j=1}^{m} T(\widehat{\alpha}_1^m + \widehat{\alpha}_2^m x_j) = 1$$
to obtain estimate of $\widehat{\alpha}_1^m, \widehat{\alpha}_2^m$.
4. For given estimate of $\widehat{\alpha}_1^m, \widehat{\alpha}_2^m$, estimate B and w by its sample analogue to obtain $\widehat{B}$ and $\widehat{w}$. Compute the estimated asymptotic variance $\widehat{v}^2 = (\widehat{w})^T(\widehat{B}^{-1})^T\Sigma\widehat{B}^{-1}\widehat{w}$.
5. The $100(1-\alpha)\%$ confidence interval for $\mathbb{E}_{\eta^{wc}}[X]$ is given by
$$I = \left(\frac{1}{m}\sum_{j=1}^{m} x_j T(\widehat{\alpha}_1^m + \widehat{\alpha}_2^m x_j) \pm \frac{\widehat{v}}{\sqrt{m}} z_{1-\alpha/2}\right).$$

Remark: The author provide a package ”divbox” in Python for direct implementation.

Review of Common Distributions

To show the usefulness of the toolkit, the explicit divergence for Weibull and Lognormal Models are shown. Recall that

- **Weibull distribution:** For $X \sim \text{Weibull}(k, \lambda)$, its density is given by

$$f(x) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, \quad \mathcal{S} = [0, \infty),$$

  which is commonly encountered in Survival Analysis, Extreme Value theory and Weather forecasting.

- **Generalized Lognormal distribution:** For $X \sim \text{Lognormal}(\mu, \sigma^2, r)$ with $r \geq 2$, its density is given by

$$\frac{1}{Zx}\exp\left\{-\frac{|\ln x - \mu|^2}{r\sigma^r}\right\}, \quad \mathcal{S} = [0, \infty),$$

  with $Z = 2r^{1/r}\sigma\Gamma(1 + 1/r)$. It is commonly encountered in standard models for losses under Solvency II and Basel III.

We would then construct the corresponding F for each case.

The Weibull Case

Weibull Assumption

$\mathcal{S} = [0, \infty)$ and $\exists k > 0$ s.t. $\varphi \sim x^k$ (Asy Equiv), where $\varphi = -\log(f)$.

Remark: Weibull Assumption is valid for (but not limited to) whole Weibull family.

Lemma (Weibull-Divergence)

Suppose the reference distribution f satisfies Weibull assumption. For $\theta > \max\{k, 1\}$, define

$$H(y) := \begin{cases} \frac{k}{\theta}\left((y+1)^{\theta/k} - 1\right) & , \text{ for } y \geq 0 \\ y & , \text{ for } y < 0 \end{cases} \quad ; H^{-1}(x) = \begin{cases} \left(\frac{\theta x}{k} + 1\right)^{k/\theta} - 1 & , \text{ for } x \geq 0 \\ x & , \text{ for } x < 0 \end{cases}$$

Then proposition (1)-(3) is satisfied and hence $\mathcal{D}_F = \mathcal{D}_F(\theta)$. Furthermore, the worst-case density $g^{wc}(x) = h^{wc}(x) f(x)$, where

$$h^{wc}(x) = \begin{cases} \exp\left\{\left(\frac{\theta}{k}(\alpha_1^{wc} + \alpha_2^{wc} x) + 1\right)^{k/\theta} - 1\right\} & , \text{ for } \alpha_1^{wc} + \alpha_2^{wc} x \geq 0 \\ \exp(\alpha_1^{wc} + \alpha_2^{wc} x) & , \text{ for } \alpha_1^{wc} + \alpha_2^{wc} x < 0 \end{cases}$$

for some $\alpha_1^{wc} \in \mathbb{R}, \alpha_2^{wc} \in \mathbb{R}^+$. The function h is called as the wors-case change of measure.

The Generalized Lognormal Case

Lemma (Generalized log-normal Divergence)

For the reference distribution f being Lognormal(μ, σ^2, r) and $\theta > 1$, define its associated H as

$$H(y) = \begin{cases} \frac{1}{(\theta\sigma)^r}\left(\exp\left\{\left(r(\theta\sigma)^r y + 1\right)^{1/r} - 1\right\} - 1\right) & \text{, for } y \geq 0 \\ y & \text{, for } y < 0 \end{cases}$$

with its inverse H^{-1} being

$$H^{-1}(x) = \begin{cases} \frac{1}{r(\theta\sigma)^r}\left(\left[\log\left((\theta\sigma)^r x + 1\right) + 1\right]^r - 1\right) & \text{, for } x \geq 0 \\ x & \text{, for } x < 0 \end{cases}$$

Then proposition (1)-(3) is satisfied and hence $\mathcal{D}_F = \mathcal{D}_F(\theta)$. Furthermore, the worst-case density $g^{wc}(x) = h^{wc}(x)f(x)$, where for $y = \alpha_1^{wc} + \alpha_2^{wc}x$,

$$h^{wc}(x) = \begin{cases} \exp\left\{\frac{1}{r(\theta\sigma)^r}\left[\log\left((\theta\sigma)^r(\alpha_1^{wc} + \alpha_2^{wc}x) + 1\right) + 1\right]^r - 1\right\} & \text{, for } y \geq 0 \\ \exp(\alpha_1^{wc} + \alpha_2^{wc}x) & \text{, for } y < 0 \end{cases}$$

for some $\alpha_1^{wc} \in \mathbb{R}, \alpha_2^{wc} \in \mathbb{R}^+$.

Agenda

Sufficient Conditions for Multivariate Problems

Consider $G : \mathbb{R}^n \to \mathbb{R}$ and define the random vector $Z = (Z_1, \cdots, Z_n)$.

Motivation

In many application, we might be interested in $X = G(Z)$. Suppose we only know marginal distribution of each Z_i, then how can us comment on

$$\sup_{D_F(\eta|v)<\kappa} \mathbb{E}_\eta(X)$$

The main difficulties are the following:

- The dependence among Z_i's may not be known.
- Even we know the joint distribution of Z, the verification of condition in Proposition 1, which involving integral associated to f, is extremely difficult.

Sufficient Conditions for Multivariate Problems

Due to the mentioned difficulties, it is hard to verify whether $\sup_{D_F(\eta|v)<\kappa} \mathbb{E}_\eta(X) < \infty$. The author propose the following remedy:

Proposition 4

Suppose H satisfies all the regularity condition mentioned before. Then for reference models v and $v_1, \cdots, v_n$ with $X \sim v$ and $Z_i \sim v_i$, assume $\exists$ constant $C > 0$ s.t.

$$|X| \leq C \left(1 + \sum_{i=1}^{n} |Z_i| \right)$$

almost surely, and all v_i satisfies condition in Proposition 1. If $x \mapsto x \exp(H^{-1}(x))$ is convex, then v also satisfies condition in Proposition 1, i.e. $\sup_{D_F(\eta|v)<\kappa} \mathbb{E}_\eta(X) < \infty$.

Comments on the Proposition

- It breaks down the difficult problem (X) into less difficult problems (Z_i's).
- It works without any assumption on dependence structure of the Z_i's.

Equivalence of Univariate and Multivariate Worst-Case

Consider the setting below

Question

For $X = G(Z)$, where $Z = (Z_1, \cdots, Z_n)$ with support $\mathcal{U}$ and $G : \mathbb{R}^n \to \mathbb{R}$,

1. Joint Distribution of Z under reference model, v_Z.
2. Marginal Distribution of $X = G(Z)$ under reference model, v.

Intuitively speaking, the first one contain more "complete" information. Then should we replace the worst-case scenario by its multivariate analogue, i.e.

$$\sup_{\eta_Z : D_F(\eta_Z | v_Z) \leq \kappa} \mathbb{E}_{\eta_Z}[G(Z)], \quad \text{where} \quad D_F(\eta_Z | v_Z) = \int_{\mathcal{U}} F\left(\frac{g_Z(z)}{f_Z(z)}\right) f_Z(z) dz$$

Then should we do so?

Fortunately, it turns out that the univariate case suffices for our study.

Proposition 5

The univariate and multivariate worst-case are equivalent, i.e.

$$\boxed{\sup_{\eta : D_F(\eta|v) \leq \kappa} \mathbb{E}_\eta[X]} = \boxed{\sup_{\eta_Z : D_F(\eta_Z|v_Z) \leq \kappa} \mathbb{E}_{\eta_Z}[G(Z)]}$$

Assume the above quantities are finite, we have

1. Closed form Multivariate Worst-Case density:

$$\boxed{g^{wc}(x) = h^{wc}(x)f(x)} \Rightarrow \boxed{g_Z^{wc}(z) = h^{wc}(G(z))f_Z(z)}$$

2. $\boxed{g_Z^{wc}(z) = h_Z^{wc}(z)f_Z(z)} \Rightarrow \boxed{g^{wc}(x) = \mathbb{E}_{v_Z}[h_Z^{wc}(Z)|G(Z) = x]f(x)}$

Proposition 5 allow as to investigate a broader class of problem, for example

Example 4 (Best-Case Analysis)

Define best case of X in infimum sense. We have best-case problem for X as

$$\inf_{\eta : D_F(\eta|v) \leq \kappa} \mathbb{E}_\eta[X] = -\sup_{\eta : D_F(\eta|v) \leq \kappa} \mathbb{E}_\eta[-X] \overset{\text{Prop 5}}{=} -\sup_{\bar{\eta} : D_F(\bar{\eta}|\bar{v}) \leq \kappa} \mathbb{E}_{\bar{\eta}}[Z],$$

where $Z = G(X) = -X$ with $\bar{v}$ as distribution of Z.

Comparison with the Wasserstein Approach

Wasserstein approach is the main competitor of F-divergence to model uncertainty:

Definition (Wasserstein distance)

For two distributions μ_1 and μ_2 on $\mathbb{R}^n$, the Wasserstein distance d_W is given by

$$d_W(\mu_1, \mu_2) = \inf_{(X_1, X_2) \sim \mu \in M(\mu_1, \mu_2)} \mathbb{E}_\mu |X_1 - X_2|,$$

where $|\cdot|$ is the Euclidean norm and the infimum is taken over set $M(\mu_1, \mu_2)$ of all coupling of μ_1, μ_2.

The Wasserstein analogues of univariate and multivariate worst-case problems are

$$\sup_{\eta : d_W(\eta, \upsilon) \leq \kappa} \mathbb{E}_\eta[X] \quad \text{and} \quad \sup_{\eta_Z : d_W(\eta_Z, \upsilon_Z) \leq \kappa} \mathbb{E}_{\eta_Z}[G(Z)]$$

respectively. However, those two are not equivalent and hence this approach does not respect the structure of the problem. It is one of the main reason that people might use F-divergence approach instead.

Agenda

Example 1: Inventory Pooling

Assume a firm sells an identical good at n locations. Assume that

- D_i, the demand at location i are independent across location.
- $D_i = a\exp(bZ_i)$, i.e. D_i follows lognormal distribution.
- Costs for each unit when demands exceeds inventory: $\$p$.
- Costs for each unit when inventory exceeds demands: $\$h$.

Then the cost under fully decentralized inventory scheme, where each location had inventory amount q is given by

$$C_d(q) = h\sum_{i=1}^{n}\max(q - D_i, 0) + p\sum_{i=1}^{n}\max(D_i - q, 0).$$

For centralized inventory scheme, the costs is

$$C_c(Q) = h\max\left(Q - \sum_{i=1}^{n} D_i, 0\right) + p\max\left(\sum_{i=1}^{n} D_i - Q, 0\right).$$

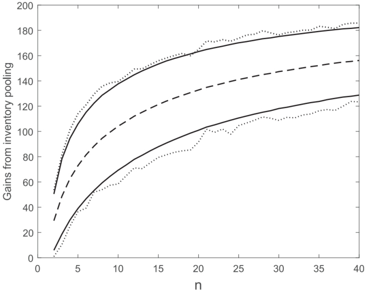
Let q_v^* and Q_v^* be minimizer of $\mathbb{E}_v(C_d(q))$ and $\mathbb{E}_v(C_c(Q))$ respectively. We then define the quantity of interest, Nominal gains per location from inventory pooling as $\mathbb{E}_v(X)$, where

$$X = \frac{1}{n}\Big(C_d(q_v^*) - C_c(Q_v^*)\Big).$$

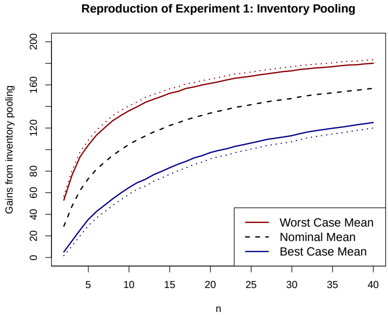
Simulation Result

Set $a = 100, b = 2^{1/2}, p = q = 1$. Also, $r = 2, \theta = 2, \kappa = 0.1$ and $\sigma = b$.

- (Dashed line) Nominal mean obtained by standard algebra.
- (Solid line) Worst and Best cases expectation: For each n, use Monte Carlo approach with 500,000 simulations of X.
- (Dotted line) Confidence bounds based on sample size $n = 1000$



(a) Simulation Result



(b) Replicated Simulation

Takeaway: The inventory pooling is worse even for minor model misspecification.

Review of Copula (Optional)

Question: How to model of joint dependence between RVs?

Given $X \sim F_X$ and $Y \sim F_Y$. Suppose we want to maintain their marginal structure, while mimicking the dependence between $U \sim F_U$ and $V \sim F_V$.

The construction can be done in following way:

1. Desired Dependence on $\text{Supp}_{U\times V}$: $(U, V) \sim F_{U,V}$
2. Desired Dependence on $[0, 1]^2$: $(F_U(U), F_V(V))$
3. Mimick Dependence on $[0, 1]^2$: $(U_X, U_Y) \coloneqq (F_U(U), F_V(V))$
4. Modelling of (X, Y) on $\text{Supp}_{X\times Y}$: $(X, Y) \coloneqq (F_X^{-1}(U_X), F_Y^{-1}(U_Y))$

In short, Copula is a tool to give a desired dependence in step (2).

Definition (Copula)

$C = C(u_1, \cdots, u_n)$ is a Copula if C is a joint-distribution of $U_1, \cdots, U_n \sim \text{Unif}[0, 1]$.

Examples of Copula (Optional)

We give examples of some commonly used copula:

Example 5 (Gaussian coupla and t-copula)

- Gaussian copula: $C_G(u_1, \cdots, u_n) = \mathbf{\Phi}_\Sigma(\Phi^{-1}(u_1), \cdots, \Phi^{-1}(u_n))$, where $\mathbf{\Phi}_\Sigma(\cdot)$ and $\Phi(\cdot)$ is the CDF of $\mathcal{N}_n(\mathbf{0}, \mathbf{\Sigma})$ and $N(0,1)$ respectively.
- t-copula: $C_t(u_1, \cdots, u_n) = T_{v,R}(t_v^{-1}(u_1), \cdots, t_v^{-1}(u_n))$, where $T_{v,R}$ is the CDF of multivariate t-distribution with d.f. v and correlation matrix R. While t_v is the univariate t-distribution with d.f. v.

Noticing that t-distribution is heavy-tailed and hence it exhibits tail-dependence.

Example 2: Proportional Reinsurance

We consider the following setting:

- Reinsurer provides protection for fixed proportion $\rho \in [0,1]$ of total claim.
- Individual claims: $L_i \sim$ Weibull(k_i, λ_i) marginally.

Our quantity of interest is thus

$$\boxed{R = \rho \sum_{i=1}^{n} L_i \quad \Rightarrow \quad \sup_{D_F(\eta|v) \leq \kappa} \mathbb{E}_\eta[R]}$$

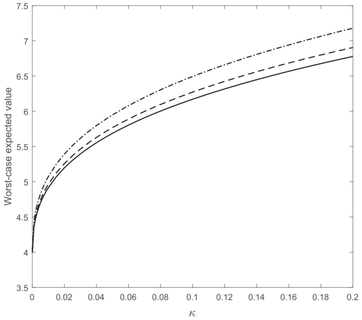
- F is chosen as in the Weibull Case with parameter $(\theta = 2, k = 0.3)$, i.e. the induced uncertainty set only contain distribution in $\mathcal{L}^2$.
- $n = 5$, $(k_1, \cdots, k_5) = (0.3, 0.4, 0.5, 0.6, 0.7)$, $\lambda_i = 1.1^{-1/k_i}$, $\rho = 0.3$
- The degree of freedom $v = 2$ and the correlation matrix is

$$\Sigma = \begin{pmatrix} 1 & -0.25 & 0.25 & 0 & 0.25 \\ -0.25 & 1 & 0 & 0.25 & 0.25 \\ 0.25 & 0 & 1 & 0 & 0.25 \\ 0 & 0.25 & 0 & 1 & 0.5 \\ 0.25 & 0.25 & 0.25 & 0.5 & 1 \end{pmatrix}$$

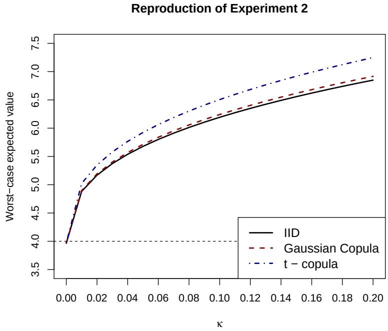
Simulation Result

We consider the following three reference model for joint distribution of L_i:

1. (Solid Line) $(L_1, \cdots, L_5)$ are independent.
2. (Dashed Line) $(L_1, \cdots, L_5)$ follows Gaussian copula with correlation matrix Σ.
3. (Dashed-Dotted Line) $(L_1, \cdots, L_5)$ follows t-copula with correlation matrix Σ and degree of freedom $v = 2$.



(a) Simulation Result



(b) Replicated Simulation

Takeaway: The t-copula, which accounts for tail dependence, yields markedly higher worst-case expected values for any given radius.

Agenda

Conclusion

To summarize, the author proposed a toolkit so that we could study the quantity

$$W(v, F, \kappa) = \sup_{\eta : D_F(\eta|v) \leq \kappa} E_\eta(X)$$

better in the following sense: It provide a new parameterization of $\boldsymbol{F}$-divergence so that we can have the following nice properties:

- Way to ensure finiteness of $W(v, F, \kappa)$.
- Closed-form expression of worst-case density g^{wc} for further analysis.
- Construction guideline to tailor made F so that we can ensure content of the uncertainty set $\{\eta : D_F(\eta|v) \leq \kappa\}$
- Easier point and interval estimation of W in followsing sense:
  1. Easier Numerical Optimization due to existence of closed form g^{wc} and H^{-1}.
  2. More theoretical guarantee based on the Method of Moment estimator.

which provide many useful real life applications. However, the toolkit have far more kind of applications and could be extended further. We are going to demonstrate them in later sections.

Agenda

Agenda

Review of the Background

Suppose we are going to test some general hypothesis $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$. Commonly Frequentist Statistician consider test statistics in form of

$$T = T(X_1, \cdots, X_n) = \frac{\Delta(X_1, \cdots, X_n)}{\widehat{\sigma}(\Delta(X_1, \cdots, X_n))},$$

and under suitable regularity conditions, we commonly have $T \xrightarrow{d} N(0, 1)$ under H_0.

- Δ is the main term of test statistics, i.e. to detect the discrepancy from H_0.
- $\widehat{\sigma}$ is a consistent estimator of standard deviation of the main term, serving as the standardizer of the test such that $T \xrightarrow{d} N(0, 1)$ under H_0.

We reject H_0 when value of T is extreme, eg: $|T| \geq z_{1-\alpha/2}$, where α is size of the test. Under H_1, we do not have $T \sim N(0, 1)$ in general. We should construct T such that

Criteria for a good Test Statistics

- Δ reflect discrepancy from H_0 **(Different value under H_0 and H_1)**
- $\widehat{\sigma}$ act as only a standardizer **(Similar value under H_0 and H_1)**

Motivation and Partial solution to the problem

Suppose one propose a model η on $(X_1, \cdots, X_n)$. Indeed, it is common to encounter the following phenomenon: As the discrepancy from H_0 increases,

$$\boxed{|\Delta| \uparrow \quad \text{and} \quad \widehat{\sigma} \uparrow\uparrow} \quad \text{such that} \quad \boxed{|T| = |\Delta / \widehat{\sigma}| \downarrow} \quad \Rightarrow \quad \boxed{H_0 \text{ is NOT rejected}}$$

i.e. the $\widehat{\sigma}$ is unstable in sense that it grows in faster rate than $|\Delta|$.

Takeaway: The stability of $\widehat{\sigma}$ is important for constructing good test statistics.

Therefore, it motivates to estimate the value of

$$\boxed{\delta(\widehat{\sigma}) := \frac{\sup_{\eta : D_F(\eta | v) \leq \kappa} \mathbb{E}_\eta(\widehat{\sigma}) - \mathbb{E}_v(\widehat{\sigma})}{\mathbb{E}_v(\widehat{\sigma})}}$$

as a measure of the stability of the standardizer and hence the quality of the test statistics. $\widehat{\sigma}$ is good if $\delta(\widehat{\sigma})$ is closed to 0. The task could be done by the toolkit if

1. $\exists C > 0$ s.t. $\widehat{\sigma} \leq C(1 + \sum_{i=1}^n |X_i|)$ almost surely.
2. $X_i \sim v_i$ and v_i satisfies $\int_{\mathcal{S}} |x_i| \exp\left(H^{-1}(\alpha |x_i|)\right) f_{X_i}(x_i) dx_i < \infty$ for $\alpha \in \mathbb{R}^+$.
3. The mapping $x \mapsto x \exp(H^{-1}(x))$ is convex.

Illustrative Example for New Application

Let $X_1, \cdots, X_n \overset{iid}{\sim} f$ with measure v and $\mu := \mathbb{E}_v(X)$, where X_i are discrete non-negative valued R.V. Suppose we want to test

$$H_0 : \ \mu = \mu_0 \quad \text{vs} \quad H_1 : \ \mu \neq \mu_0$$

Different people may propose different test statistics according to their principle. Say

- Student A (Parametric Guy): He/She believe that f is indeed Poisson, and hence under H_0, $\mathbb{E}(X) = \mathbb{V}ar(X) = \mu$ and proposed the following test statistics

$$T_A = \frac{\bar{X}_n - \mu_0}{\sqrt{\bar{X}_n / n}}$$

- Student B (Non-parametric Guy): He/She does not want to put too many distributional assume f and hence proposed

$$T_B = \frac{\bar{X}_n - \mu_0}{\sqrt{n^{-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2}}$$

Notice that T_A and T_B only differ in the standardizer, write

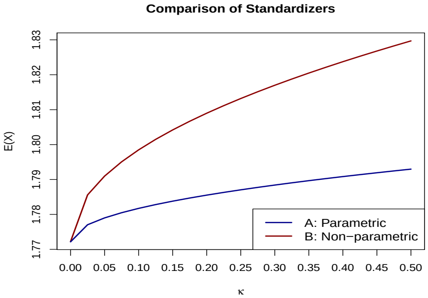
- $\widehat{\sigma}_A^2 = \bar{X}_n / n$
- $\widehat{\sigma}_B^2 = \sum_{i=1}^{n} (X_i - \bar{X}_n)^2 / n$

Simulation Studies

Important Remark

It is important to ensure $\widehat{\sigma}_A \xrightarrow{pr} \sigma$ and $\widehat{\sigma}_B \xrightarrow{pr} \sigma$ under H_0, where $\sigma^2 = \mathbb{V}ar_v(\Delta)$. Otherwise, it is NOT a fair comparison.

The task reduced to worst-case study of these two quantities. In the simulation, we assume that $X_1, \cdots, X_n \overset{iid}{\sim}$ Po(π). The simulation result is as follow:



The result is consistent to our intuition, for all κ, we have $\delta(\widehat{\sigma}_A) < \delta(\widehat{\sigma}_B)$ as the standardizer in T_A utilize the distributional assumption of f.

Agenda

Background and Motivation

Let T_A and T_B be test statistics for some prespecified H_0 vs H_1. In general framework of Hypothesis Testing, we commonly encounter the following situation:

1. ($\Delta_A \neq \Delta_B$) Test statistics with different main term. Hence it is inappropriate to conduct stability test on $\widehat{\sigma}_A$ and $\widehat{\sigma}_B$.
2. (Complicated H_0 & H_1) The H_0 and H_1 may not be parameterizable, hence hard to give a general measure of discrepancy for constructing power curve.

Consider the **distribution testing problem** on $X_1, \cdots, X_n$ as an example:

$$H_0 : \ v = v_0 \quad \text{vs} \quad H_1 : \ v \neq v_0$$

where v and v_0 are the true and proposed distribution of X_i's respectively. Obviously ,

1. It is impossible to parameterize H_0 or H_1.
2. There are no unique measure of discrepancy between H_0 and H_1.

Generalized Power function

Recall that test-statistics is a function of the RVs, i.e. $T = T(X_1, \cdots, X_n)$ and after theoretical justification, one would decide the rejection region R such that

$$H_0 \text{ is rejected} \iff T \in R \iff \widehat{\psi} := \mathbb{1}(T \in R) = 1$$

Therefore, $\mathbb{E}_v(\widehat{\psi}) = \mathbb{P}_v(T \in R) = \mathbb{P}_v(H_0 \text{ is rejected})$, where v is the reference model. It motivates us to propose the following:

Definition (Generalized Power Function)

For $T = T(X_1, \cdots, X_n)$ under reference model, we define the generalized power function associated to F as

$$\beta_{\widehat{\psi}}(\kappa) := \sup_{\eta : D_F(\eta | v) \leq \kappa} \mathbb{E}_\eta(\widehat{\psi}),$$

the interpretation to it is exactly same as the standard power curve. In additional,

1. It utilize the F-divergence as the measure of discrepancy from H_0.
2. We need NOT to generate T under H_1 to obtain estimate of $\beta_{\widehat{\psi}}(\kappa)$.
3. $\beta_{\widehat{\psi}}$ is monotonically increasing function in κ according to definition.

Illustrative Example

We give an explicit example here: Suppose $X_1, \cdots, X_n \overset{iid}{\sim} v$ and we propose v_0 as the CDF for the Gaussian Mixture $I * Z_1 + (1 - I) * Z_2$, where $I \sim$ Bern$(0.3), Z_1 \sim N(0,1), Z_2 \sim N(1, 2^2)$. Then we test

$$H_0 : v = v_0 \quad \text{vs} \quad H_1 : v \neq v_0$$

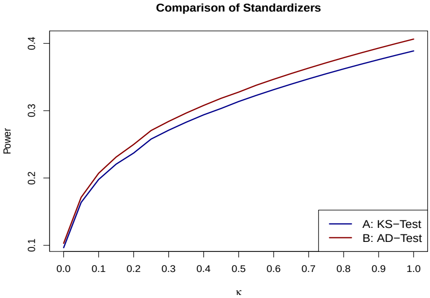
with two different test statistics:

1. (Kolmogorov-Smirnov test): $K_n = \sqrt{n} \sup_{x \in \mathbb{R}} |\widehat{v}(x) - v_0(x)|$.
2. (Andersen-Darling test): $A_n = n \int_{-\infty}^{\infty} \frac{\left(\widehat{v}(x) - v_0(x)\right)^2}{v_0(x)(1 - v_0(x))} dv_0(x)$.

Commonly, people proposed different kind of measure of discrepancy from H_0 to construct the power curve, for example, let δ be the mean shift/variance shift from H_0. However, it **exclude many distribution in neighbourhood of H_0**.
Instead, we compare those two tests by the Generalized Power Function.

Simulation Studies and Comments

The following is the simulation result:



which is consistent to the ordinary result that AD-test is commonly more powerful than KS-test. We then make some comment on the proposal.

- 👍 Provide a new and general measure of discrepancy from H_0.
- 👍 Generation of replicates under H_0 are sufficient to evaluate the whole function.
- 👎 Careful selection of the F is necessary to ensure finiteness of the estimator.
- 👎 There is hidden moment restriction on H_1 depending on choice of F.

Agenda

Agenda

Motivation

Consider the multivariate worst-case problem of $X = G(Z_1, \cdots, Z_n)$, where $Z_1, \cdots, Z_n$ are iid. Throughout the whole paper and the simulation studies. we are assumed to be capable to

1. Specify a reference model η to X.
2. Generate replicates of X or $(Z_1, \cdots, Z_n)$ to estimate the worst case density.

However in reality,

1. We may not be able to propose a reasonable model due to the complicated structure of G or Z_i's.
2. The distribution of X or Z_i's might be complicated (or even unknown), which leads to difficulties in generating replicates.
3. Commonly, only **single realization** of $Z_1, \cdots, Z_n$ is available.

It motivates us to consider the following question:

If we are only given a single realization of $\boldsymbol{Z} = (Z_1, \cdots, Z_n)$, where $Z_1, \cdots, Z_n$ are iid. Then how can we estimate

$$\sup_{\eta : D_F(\eta|v) \le \kappa} \mathbb{E}[X], \quad \text{where } X = G(\boldsymbol{Z})$$

by imposing minimal distributional assumption on $\boldsymbol{Z}$ or X?

Solution to the problem

To start with, we introduce the notation of bootstrap.

Definition (Bootstrap)

Suppose $Z_1, \cdots, Z_n$ are iid, then $\boldsymbol{Z}^* = (Z_1^*, \cdots, Z_n^*)$ is a **bootstrap sample** if

$$[\boldsymbol{Z}^*|\boldsymbol{Z}] = \left[Z_1^*, \cdots, Z_n^* | Z_1, \cdots, Z_n\right] \sim \text{Unif}\left\{Z_1, \cdots, Z_n\right\}$$

and the **Principle of Bootstrap** state the variability of resamples in a given sample should approximate the variability of samples in a given population.

We then utilize the bootstrap sample as replicate of $\boldsymbol{Z}$ to execute the toolkit.

Theorem (Nonparametric Worst-Case Analysis)

Suppose $Z_1, \cdots, Z_n \overset{iid}{\sim} \eta_Z$ (Without explicit form). Assume the following:

1. $\exists C > 0$ s.t. $|X| \leq C(1 + \sum_{i=1}^n |Z_i|)$ almost surely.
2. $\int_{\mathcal{S}} |z| \exp\left(H^{-1}(\alpha|z|)\right) f_Z(z)dz < \infty$ whenever $\alpha \in \mathbb{R}^+$.
3. The mapping $x \mapsto x\exp(H^{-1}(x))$ is convex.

Then we could follow the procedure in Section 5 by replacing the replicates by bootstrap sample $X_1^*, \cdots, X_M^*$ with $X_i^* = G(\boldsymbol{Z}^{*(i)})$ for large M.

Agenda

Motivation

As summarized in the previous section, the author focus on giving better understanding on $W(v, F, \kappa) = \sup_{\eta: D_F(\eta|v) \leq \kappa} E_\eta(X)$ and interpret it as

A measure of how bad the worst case could be

However, in my point of view, the methodology could be extended to

Provide a robust measure of central tendency of X

To be explicit, we want to provide a solution to the following question:

Problem to be answered

Suppose one want to estimate expected value of X and proposed model v. He/she is not very confident about his model, but at least is confident that $D_F(v^*|v) \leq \kappa$. The original question studied is that

How bad could the guess be?

while we now attempt to answer

How to estimate $\mathbb{E}(X)$ such that it works well over whole uncertainty set?

Proposal of the estimator

Definition (Divergence-Robust Estimator)

The divergence-robust estimator is constructed by the following steps:

1. **(Maximize over Uncertainty Set)** For all fixed value of θ, one can use to toolkit to estimate the maximal discrepancy

$$l(\theta) := \sup_{\eta : D_F(\eta|v) \leq \kappa} E_\eta(|X - \theta|^2)$$

   by the estimator $\widehat{l(\theta)}$.

2. **(Minimization over $\mathcal{S}$)** Define the divergence-robust estimator by

$$\widehat{\theta}_D = \arg\min_{\theta \in \mathcal{S}} \widehat{l(\theta)}$$

Interpretation of steps

- (Step 1): If we estimate the mean of X by θ, the its discrepancy with X could be as large as value of $l(\theta)$.
- (Step 2): Minimize l over $\mathcal{S}$ so that the maximal discrepancy l is minimized.

Properties of $\widehat{\theta}_D$

We first give some remark on the estimator

1. $\widehat{\theta}_D$ is a semi-parametric or non-parametric estimator as it involves both
   - Maximization over the uncertainty set, which is not a paremetrized set.
   - Modelling of reference model η, which is parametric.
2. The proposed estimator could be further generalized to estimation of $\mathbb{E}(G(X))$ for some transformation G according to the equivalence proposed by author.
3. One can modify the loss function in l by absolute value to estimate the median analogue instead.
4. The proposal is an analogue to the minimax estimator in the frequentist framework.

The main disadvantage of the estimator is that

- The minimization in second step is likely to be computationally intensive.
- It is hard to derive the asymptotic distribution of $\widehat{\theta}_D$.

Agenda

Review and Motivation

Recall that in the simulation step (1), we have to be able to compute

$$F(y) = \int_1^y H\Big(\log(z)\Big)dz$$

for all y. The author suggest that (in section 5.1.3) for realized samples $x_1, \cdots, x_n$,

1. Evaluate F on grid that covers entire range on sample values, i.e. $\mathcal{I} = [\min_i(x_i), \max_i(x_i)]$, where the maximization and minimization are taken over those positive value of x_i.
2. Partition $P : \min(x_i) = p_0 < \cdots < p_k = \max(x_i)$ of $\mathcal{I}$ and approximate the value of integral F by idea of Riemann integral to get $F(p_0), \cdots, F(p_k)$.
3. Approximate the function on $\mathcal{I}$ by spline interpolation.

The main problem arises in step (1) as the interval may be very wide, then

- Finner partition is required and hence heavier computational cost.
- Partition that is not fine enough leads to **degenerated estimate** of T, i.e. $T = 1$ and the worst-case study is not feasible.

Solution

We then provide the solution to this problem. For iid $X_1, \cdots, X_n$, define

$$Z_i = \frac{X_i - \mathbb{E}_v(X_i)}{\sqrt{\mathbb{V}ar_v(X_i)}} \quad \text{and} \quad \widehat{Z}_i = \frac{X_i - \bar{X}_n}{\sqrt{n^{-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}}$$

Noticing that for $n \to \infty$, we have $\widehat{Z}_i \xrightarrow{pr} Z_i$ under law v. It follows that

$$\begin{aligned}
\sup_{\eta: D_F(\eta|v) \le \kappa} \mathbb{E}_\eta(X) &= \mathbb{E}_v(X) + \sqrt{\mathbb{V}ar_v(X)} \sup_{\eta: D_F(\eta|v) \le \kappa} \mathbb{E}_\eta\left(\frac{X - \mathbb{E}_v(X)}{\sqrt{\mathbb{V}ar_v(X)}}\right) \\
&= \mathbb{E}_v(X) + \sqrt{\mathbb{V}ar_v(X)} \sup_{\eta: D_F(\eta|v) \le \kappa} \mathbb{E}_\eta\left(Z\right)
\end{aligned}$$

and we can regards $\widehat{Z}_i$ as Z_i to apply the toolkit. It makes the task easier as range of Z_i are restricted through standardization and we can consider the partition in a much narrower range, it can be easily justified by the Chebyshev's Inequality.

Agenda

Background

In this section, we apply the toolkit with the nonparametric reference model as proposed in the extension section to the COVID19 data set in Hong Kong. In particular, we have $X_1, \cdots, X_n$ and X_i is the number of confirmed cases of COVID19 in i-th day of 2020 in Hong Kong. It motivate us to consider worst-case study of X_i's. However, $X_1, \cdots, X_n$ is NOT iid in general, and the following problem arises

1. The worst-case expectation of each X_i are different.
2. We are NOT able to generate iid copies of each X_i's.

We tackle those problem by considering the following quantity of interest instead

$$\theta(\kappa) := \sup_{\eta : D_F(\eta|v) \leq \kappa} \mathbb{E}_\eta(\bar{X}_n)$$

so that

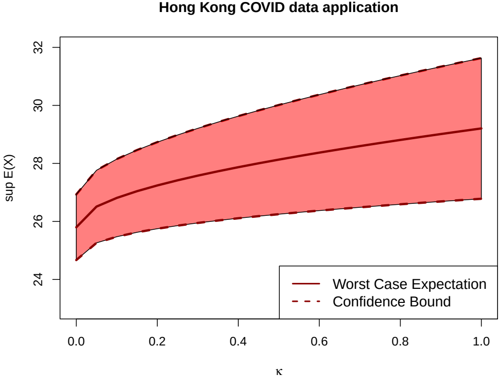
1. (Better interpretation) It fits natural interest to understand the worst case expectation over the whole year, instead of focusing on a single day.
2. The quantity could be estimated by using bootstrap replicates.

Simulation Result

The main advantage of studying the quantity θ is that

By CLT, we have asymptotic normality of $\bar{X}_n$.
$\Rightarrow$ Choose F in Weibull case with $k = 2$ and $\theta > k$ ensure the finiteness of W.

The following result is the plot of $\theta(\kappa)$ together with the confidence bound:



Remark: Click here for the source of dataset.

The END

Thank You!